Reciprocal normalization of the Gini block

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Round143 retains the stable R137 GI-TBC route, operation, inventory and selector structure. The research question is whether a second, normalized use of the same inventory selectors supplies useful information beyond the old shared-g products. The objective remains the original $g + \lambda P$. The penalty is never multiplied by the reciprocal variable. RN is an intersection with the old model, with its original integer constraints and stable cuts retained.

Exact integer extension and compact implementation

Use the actual binary64 coefficient c_i loaded in the old row $r_i = c_i Y_i$. Let $S = \sum_i c_i Y_i$ and $A = \sum_{i<j} |c_i Y_i - c_j Y_j|$. On a full model scope with $0 < S_L \le S \le S_U$, put $L_t = 1/S_U$ and $U_t = 1/S_L$, rounded outward. For the existing station states, RN adds continuous $t, w, \widehat{Y}, \widehat{h}$:

$$L_t \le t \le U_t, \quad L_t s_{iy} \le w_{iy} \le U_t s_{iy}, \quad \sum_y w_{iy} = t, \quad \widehat{Y}_i = \sum_y y w_{iy},$$

$$\sum_i c_i \widehat{Y}_i = 1, \quad \widehat{h}_{ij} \ge \pm(c_i \widehat{Y}_i - c_j \widehat{Y}_j), \quad \sum_{i<j} \widehat{h}_{ij} \le ng.$$

All new columns are continuous and nonnegative. The product facets $w \geq t - U_t(1-s)$ and $w \leq t - L_t(1-s)$ follow from the station mass equality, $\sum_y s_{iy} = 1$, and the other states' upper/lower bounds. Omitting them is therefore an exact compact implementation of RN1. No old state_g, zprod or h column is reused: they have different physical units.

At integer selectors, every unselected w is zero and the selected $w = t$. Hence $\widehat{Y} = tY$ and normalization gives $tS = 1$. The new Gini row says $g \geq A/(nS)$. Conversely, any original integer feasible point with $S > 0$ and any legal epigraph $g \geq A/(nS)$ extends by $t = 1/S$, $w = ts$, $\widehat{Y} = tY$, $\widehat{h}_{ij} = t|c_iY_i - c_jY_j|$. The original model g is retained even when it exceeds physical G; its current g interval must still be respected. This extension proof does not authorize a physical MIP Start whose true G lies outside that interval.

Thus RN preserves the original integer problem and its old-space projection is contained in the old continuous relaxation. Containment alone proves neither a strict optimum gain nor faster complete search.

The production writer adds two finite caps with the same old-space projection. $\widehat{Y}_i \leq M_iU_t$ follows from nonnegative w and $\sum_y w_{iy} = t$. Since $v_i = c_i\widehat{Y}_i \geq 0$ and $\sum_i v_i = 1$, the minimum necessary $\widehat{h}_{ij} = |v_i - v_j|$ is at most one. Any feasible RN extension can reduce each $\widehat{h}$ to that minimum without changing an old column or violating the Gini upper row; $\widehat{h}_{ij} \leq 1$ is therefore redundant in the projection. These finite caps permit the established full-matrix/Start reader to remain unchanged. They are not an extra integer-count strengthening.

Full-scope denominator bounds and numerical transport

Empty departure, loaded return and conservation imply $B^- = \max(0, B - \sum_k Q_k) \leq \tau = \sum_i Y_i \leq B$, with $B = \sum_i b_i$ excluding the depot. Cumulative pickups are not bounded by fleet capacity. For actual supported endpoints m_i, M_i, set $\tau_L = \max(B^-, \sum_i m_i)$ and $\tau_U = \min(B, \sum_i M_i)$. An incompatible interval follows the existing infeasible-domain route and is never inverted.

Start from $Y = m$. The minimum S fills $\tau_L - \sum_i m_i$ inventory units in ascending c_i order; the maximum fills $\tau_U - \sum_i m_i$ in descending c_i order. Positive coefficients make those total-inventory endpoints optimal. This continuous box-budget problem relaxes support holes and all routing constraints, so its bounds are valid for the full original scope without another Optimize or enumerating S. No observed S, incumbent inventory or heuristic sample supplies a bound.

The diagnostic oracle uses Fraction values of actual loaded binary64 c_i. Production reuses checked common-dyadic __int128 arithmetic from R140 for the complete coefficient/inventory sums. It converts S_L downward and S_U upward, then rounds each reciprocal outward. This covers accumulated summation error before inversion. Checked arithmetic fails closed outside its representable range. New row coefficients bypass the legacy near-zero/near-one formatter; an exact positive power-of-two row scale, if necessary, preserves coefficients that a Native LP parser would otherwise omit. No feasibility/optimality tolerance is loosened.

Every model uses its own actual supported interval and physical budget. A parent-valid bound would remain valid on a child, but a local bound is never exported to a larger scope. The single applicability rule is a proved positive full-scope S_L. If its lower bound is zero, RN contributes no rows or columns and the old exact model and S=0 convention remain. The optional zero/positive disjunction in the goal is not implemented: no evidence currently requires that second controller.

Strict separation and an explicit limitation

The exact three-station fixture has $c = (1, 1, 1)$, support $\{0, 1\}$, $\tau \in [1, 2]$, inventory-one selector masses $(1, 1/2, 0)$, old $g = 1/3$ and inventory-one old g-products $(1/3, 1/3, 0)$. $H = 2$, $J = 2/3$, $\tau = 3/2$ satisfy the declared old Gini, stock, TC and TC2 blocks. Endpoint pair transport gives the same absolute differences,

and the penalty on this support is linear. This is a specified relaxation subblock witness, not a fabricated full BRP instance.

RN has $t \in [1/2, 1]$, $v = (t, 1-t, 0)$. The middle site's zero-state product is $2t - 1 \geq 1/4$, so $t \geq 5/8$ and $g \geq 2t/3 \geq 5/12$. Equality is attained at $t = 5/8$ with $w = ((0, 5/8), (1/4, 3/8), (5/8, 0))$. The strict separation is exactly $1/12$.

For four stations with $c_i = 1$, support $\{0, 1\}$, fixed $\tau = 3$ and all inventory-one masses $3/4$, RN admits $t = 1/3$, $w_1 = 1/4, w_0 = 1/12$ and $v_i = 1/4$, hence $g = 0$. Old TC gives $H \geq 3$, fixed-inventory stock gives $J = 3g$, and $H \leq 4J$ implies $g \geq 1/4$. RN alone does not dominate the retained integer-count information and does not close the full R133 integer-objective mixture gap. This directly supports the additive design; no replacement-model comparison grid is warranted.

Twelve fixed-point checks and their scope

The original R140 carrier is bound by SHA256 `2cb1ab936753ce9ac26ae1f970b070b2751190af79f6a80a8b0f5d254902bf5f`. The separately named supplemental offline ZIP was not present locally. The portable checker therefore reconstructs all twelve checks from that original carrier, without loading Native or rerunning any old optimization. These are newly reconstructed certificates, not an assertion that the missing ZIP was received. Some reconstructed rank/eta supports are stronger than the rounded lower bounds stated in the goal; their actual identities are retained.

With selectors fixed, each station's attainable v_i at t is an interval $[\ell_i(t), u_i(t)]$ obtained by exact fractional knapsack. Their endpoint functions are piecewise linear. The complete feasible t interval satisfies $\sum_i \ell_i(t) \leq 1 \leq \sum_i u_i(t)$, together with all station product/mass bounds. For any permutation rank vector $a_i = 2\,\mathrm{rank}_i - n - 1$, or a convex average at ties, and η,

$$ng \geq \eta + \sum_i \min\{(a_i - \eta)\ell_i(t), (a_i - \eta)u_i(t)\}.$$

The checker minimizes this expression at every endpoint and knapsack/ normalization breakpoint of the complete t interval. Six K35/RX20-T1 saved points are excluded. Rank supports may be selected using a finite search, but the final lower bound covers the whole interval exactly; unsuccessful sample search is never an exclusion proof. For K20/RX35-T2, a constructive t and water-filled attainable v give six extensions with original columns and objective unchanged. The new columns and binary64 linear residuals are saved.

These checks interpret binary64 c and selector masses as rationals. The old complete LP points are still numerical feasible optima under the inherited policy, not rational full-matrix optimality certificates. A separated saved point need not raise the reoptimized LP optimum; an extendable saved optimum does not make RN redundant throughout the old model. The four separately optimized matching LPs test the former question directly.

Historical and literature position

R88 already developed pair-CDF/endpoint transport; R138 already established the continuous selected-resource shared-factor hull application. Neither is renamed as RN. R133's integer objective mixture issue remains a distinct limitation. Earlier rejection of coarse denominator buckets, approximate reciprocals and unproved scaling of all integer variables is not an experiment on this retained-integer finite-selector construction. The R142 count-proof continuation is closed; its CP/ALL controls are not inherited into the RN candidate. R137 AM/no-strict-child is also left unchanged.

Reciprocal normalization is standard. Charnes and Cooper (1962) introduce the linear-fractional change of variables and normalized denominator. Yue, Guillen-Gosalbez and You (2013) combine that transformation with Glover linearization for MILFP reformulations; their publisher abstract explicitly identifies those ingredients. Chen and Hooker (2023), section 4.2 and the appendix's Gini formulation, already give a normalized linear formulation of the Gini criterion. Their double-pair normalization is equivalent to A/(nS) here. The project contribution being tested is the exact connection to its existing integer inventory/route structure, complementarity with prior expressions, and complete computational value. No new general fractional programming principle, full integer hull, or journal-level improvement is claimed.

Primary sources checked on 1 November 2026:

- Charnes and Cooper, 1962, DOI, and the CMU library scan's displayed transformation on p.182.
- Yue et al., 2013, publisher abstract. Full-text access was not needed or claimed for the above attribution.
- Chen and Hooker, 2023, author manuscript, section 4.2 and Gini appendix, displayed manuscript pp.13 and 36-37; published article.

Matched LP and complete-method results

RN supplies real complementary LP information, but the two priority same-PE Seed0 pairs do not show a sufficient complete-method benefit. Retain the uniform lift as research-only and stop full expansion this round; stable R137 GI-TBC remains the paper mainline.

RN supplements the old shared-g products with a common reciprocal t, normalized inventory and normalized pair differences. It keeps the original integer route/inventory structure and original penalty. Its integer extension and continuous inclusion are exact on a proved positive-denominator scope. The separate research switch defaults off. Stable R137 remains available.

Matched continuous objectives

All four new LPs are numerical optima under unchanged 1e-6 feasibility/optimality qualification, explicit Gurobi Seed 0, Threads 1 and affinity mask 4, run serially. The original complete objective, cutoff, support, g interval and old matrix are retained. BASE gains include physical inventory information used for t bounds; TC+stock controls still show information beyond that older resource/count combination.

Role	Old expression	Old L	RN L	Gain
K35	base	0.200177662	0.228090497	0.027912835
K35	tc_stock	0.218088779	0.228090497	0.0100017179
RX20-T1	base	0.331594316	0.389761616	0.0581672999
RX20-T1	tc_stock	0.37755753	0.394121979	0.0165644487

Role / expression	Old g	RN g	Old weighted P	RN weighted P
K35/base	0.0386391834	0.0665520184	0.161538479	0.161538479
K35/tc_stock	0.0565503004	0.0665520184	0.161538479	0.161538479
RX20-T1/base	0.100712705	0.17210734	0.230881611	0.217654276
RX20-T1/tc_stock	0.14667592	0.173850507	0.230881611	0.220271472

Role / expression	Added rows	Added columns	Added nonzeros	Build / solve s
K35/base	2340	1170	7474	1.7644687 / 0.5281877
K35/tc_stock	2340	1170	7474	1.8504794 / 1.6446136
RX20-T1/base	708	354	2232	0.4122916 / 0.0574667999
RX20-T1/tc_stock	708	354	2232	0.4220257 / 0.0796056001

Actual denominator bounds, reciprocal endpoints, complete row/column/nonzero counts, Runtime/Work, all quality measures and actual source/solution hashes are in reports/LP.json and LP.csv. These reoptimized full-objective gains are separate from saved-point exclusions. Four diagnostic Optimize calls were sufficient; no RN x TC2 x CP grid was opened.

Complete same-PE methods

Each method pays for its own H (separate fixed seed 20260626), root/lookahead, online validation and complete Native proof search. The stable carrier is M-B/VD-P, pair-only, AM=.08, depth 8, minimum width 1e-4, AP off; stock, TC/TC2, count disjunction and CP are off. Each entered role receives its original full cap. Diagnostic/historical solutions or bounds are never injected.

Role	Method	U	Supported L	Relative gap	T cert s
K35	stable-off	0.692854359	0.557158766	0.195850096	open
K35	RN	0.681258498	0.542931774	0.203045869	open
RX20-T1	stable-off	0.64532503	0.64532503	3.44e-16	22.206689
RX20-T1	RN	0.64532503	0.64532503	1.24e-09	23.4901348

Role / method	Produce s	Verify s	Total s	Splits
K35/stable-off	2371.14246	4.1181738	2375.26063	0
K35/RN	2371.10851	6.0289586	2377.13747	0
RX20-T1/stable-off	22.206689	1.1920446	23.3987336	0
RX20-T1/RN	23.4901348	1.4364241	24.9265589	0

Same-PE role	RN - off U	RN - off L	Produce delta s	Total delta s
K35	-0.0115958603	-0.0142269917	-0.033952195	1.87683261
RX20-T1	0	-7.97e-10	1.28344575	1.52782525

K35 is a real tradeoff: RN lowers verified U by 0.0115958602505 but also lowers supported L by 0.0142269916815. Its signed gap grows from 0.135695592698 to 0.138326724129, and relative gap grows from 0.195850096067 to 0.203045869458. Both remain qualified-open after the full 2400-second caps. Producer time is essentially unchanged; RN adds 1.91078480007 seconds of verification and 1.87683260511 seconds to complete total. RX20-T1 certifies the same verified physical objective 0.645325030386 with both methods; RN raises producer/certificate time from 22.2066890239 to 23.4901347759 seconds and total from 23.3987336238 to 24.9265588757 seconds. The 7.97e-10 returned-L difference is within the inherited numerical policy and is not a substantive mathematical loss or gain.

Open T_cert is null, never the cap. Native returned ObjVal/ObjBoundC and externally merged supported L are separately retained. Root LP, AM decisions, outer splits, terminal nodes/Work and full Native logs explain observed outcomes without equating LP strength with search speed. One Seed does not identify a runtime distribution or attribute every difference directly to a new row.

All four formal positions have zero outer splits. Actual complete root bounds repeat the matched diagnostic improvements, but final Native cut/search behavior is different. K35 terminal nodes decrease from 16825 to 10779 while Work increases from 4691.00560053 to 4999.12908278. RX20-T1 nodes decrease from 2621 to 1620 while Work increases from 29.5418051573 to 34.9746070207. These are descriptive full-method effects, not a proof that fewer nodes mean less computation or that a specific added row caused every difference. Same-PE methods have matching complete own-H semantics and fleets; their full Native/H/Start/quality/lifecycle evidence is qualified.

Qualifications, boundaries and cost

The three-station exact subblock separates 1/3 from 5/12; the four-station fixed-denominator example allows RN g=0 while retained TC+stock requires 1/4. RN is complementary and is not claimed to dominate the old count information or provide the complete integer objective hull. The original R140 carrier supplied twelve fixed-point checks with zero Optimize: six exclusions and six same-old-point extensions. The separately named offline ZIP was absent; reconstructed rank/eta certificates have their own identities and sometimes stronger bounds. They are not retroactively labeled as the missing old certificate bytes.

Production qualification includes exact dyadic full-scope bounds with directed conversion before reciprocal, actual c_i, zero-lower-bound omission, fixed-S and positive-S fixtures, nonminimal epigraph extensions, complete old/new Start values, preservation of the physical-G Start-domain rule, and actual Native reads of all new rows/coefficients/column types/bounds. K35 RN-off equals the inherited complete BASE matrix. All four affected units were recompiled; hybrid object/archive lineage and actual PE/DLL hashes are saved. Two new diagnostic failures occurred before Optimize, were repaired under new labels, and remain charged. Small offline constructor/search failures are separately recorded.

The conservative new experimental charge is **4880.559825 seconds (1.355711 hours)** of the 28800-second ceiling. This is the complete outer including qualification, failures and mandatory postverification, charged once. Nested H/LP/MIP/Work are nonadditive details. Build time is separately 37.893495 seconds; exact offline reconstruction, writing, independent review, PDF and public I/O are separate. Interactive editing/review does not receive invented timing precision.

The decision is RESEARCH_ONLY_STOP_FULL_EXPANSION_THIS_ROUND. Qualification and integer equivalence pass; nonpromotion follows the observed complete-method tradeoff and lack of proof-clock benefit, not a correctness failure. The code remains a separate uniform default-off research option. The tested result is not that RN is mathematically redundant, generally slower, or incapable of benefiting an untested instance.

No formal K20/RB35-T2 pair, optional K15/RX35-T2 pair or Seed1 is entered. K20 cap90 remains health-only; its saved-root extension is not a complete performance prediction. Other roles could characterize additional behavior, but there is no specific pending adoption/stop decision that their measurement would change after the priority hard-role support-bound regression and the second role's proof-clock result. Seed1 would not erase the retained adverse Seed0 result or establish a runtime distribution. Reopening requires a concrete new mechanism or decision-changing evidence, rather than a missing table. Stopping occurs with substantial budget remaining.

Reproduction and delivery identity

Actual production source is `85bd2304866942fd5b397970c5ef4099bb1b73fb`; PE SHA256 is `eaa6dc5d0091a1f3a38a0374aeccab28e94bff94606339c62b385b48cb7d44b1`. DLL SHA256 is `9b5fccef82043cc9a0f052d31fa9c12b6d59cec999ad0cc19ca1b68ed6d18a88`. The carrier is stable R137 plus uniform RN, not R138-R142 mechanism stacking. Source/PE/helper/input/parameter matrices and actual returned calls are bound separately from later report metadata. The research entry is `scripts/paper_s2_rn.py`, profile

`config/gi_tbc_rn_round143.json`, with `--rn on|off`; it expands only unless `--run` is explicit. `RESUME.md` describes safe continuation and the no-repeat rule.

The mathematical/history/literature note and independent reviews separate four evidence levels: standard construction and integer equivalence, strict subblock/saved-point relations, reoptimized LP gains, and complete algorithm performance. Existing R88/R133/R138/R140 results and Charnes-Cooper, Yue et al. and Chen-Hooker antecedents retain their identities. No new general fractional-programming principle or higher-journal claim follows.

The formal stable GI-TBC manuscript is retained. The increment and full reports retain all entered favorable/adverse positions; missing configurations are not inferred. Fixed-science carriers, anonymous public byte/member recovery and bounded current-round cleanup have separate receipts. All new assets stay under ExactEBRP-work, protecting the original user tree and old rounds. No merge or force push is authorized.

The standard antecedents are (Charnes and Cooper 1962; Yue et al. 2013; Chen and Hooker 2023).

References

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